THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Algebra Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $2x + 5$	2 times x is $2x$; add 5
2	**(a)** 19	$3 \times 5 + 4$
3	**(b)** $6x + 8$	Add like terms
4	**(d)** $10x + 2$	Place value: tens × 10 + ones
5	**(c)** 28	$3 + 4 + 10 + 11$
6	**(c)** 5	Number multiplying the variable
7	**(a)** $3x + 4$	3 times x is $3x$; add 4
8	**(b)** 19	$2 \times 6 + 7$
9	**(b)** $8x + 8$	Add like terms
10	**(d)** $10x + 3$	Place value: tens × 10 + ones

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$5n + 7$ (2).	2
12	$3 \times 4 + 2 \times 5 = 22$ (1+1).	1+1
13	$7x + 7y$ (1+1).	1+1
14	$4y - 9$ (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$3 \times 6 + 4 \times 2 - 5$ (2) $= 21$ (1).	2+1
16	Like terms: $9x - 5x = 4x$ (1); constants $4 + 8 - 2 = 10$ (1); result $4x + 10$ (1).	1+1+1
17	$23 + 32 = 55$ (2); $= 11 \times 5$, a multiple of 11 (1).	2+1
	OR: $5 \times 4 + 2 \times 7 - 8$ (2) $= 26$ (1).	2+1
18	Like terms: $12x - 7x = 5x$ (1); constants $3 + 6 - 4 = 5$ (1); result $5x + 5$ (1).	1+1+1
	OR: $45 + 54 = 99$ (2); $= 11 \times 9$, a multiple of 11 (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$3n + 5 - 5 = 3n$ (3); for $n = 4$, result $= 12$ (2).	3+2
	OR: Middle row: 2+5=7 (1), 5+3=8 (1); top $= 7 + 8 = 15$ (3).	1+1+3
20	$4n + 7 - 7 = 4n$ (3); for $n = 6$, result $= 24$ (2).	3+2
	OR: Middle row: 4+1=5 (1), 1+6=7 (1); top $= 5 + 7 = 12$ (3).	1+1+3

Mistakes to avoid:

- Adding unlike terms: $3x + 4x^2 = 7x^3$; only like terms are combined, and $3x + 4x = 7x$ (not $7x^2$).
- Writing the two-digit number as ab (product) instead of $10a + b$.
- Checking a trick or identity with one example only and calling it a proof; use a variable for the general case.
- Sign mistakes when removing brackets: $-(x - 3) = -x + 3$, not $-x - 3$.
- Writing $(a + b)^2 = a^2 + b^2$ or $(a - b)^2 = a^2 - b^2$ while simplifying expressions.
- Deriving a pattern rule from the first term only; check that the rule works for every given term (e.g. $3n + 1$ for $n = 1, 2, 3$).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Algebra Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 40	$6+7+13+14$
2	**(c)** $3x$ and $9x$	Same variable and power
3	**(c)** $5x+7$	5 times x is $5x$; add 7
4	**(a)** 23	$5\times4+3$
5	**(c)** $9x+10$	Add like terms
6	**(c)** $10x+4$	Place value: tens × 10 + ones
7	**(a)** 52	$9+10+16+17$
8	**(a)** 3	Terms separated by + or −
9	**(d)** $4x+9$	4 times x is $4x$; add 9
10	**(b)** 21	$4\times3+9$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$5\times2+4\times6=34$ (1+1).	1+1
12	$8x+9y$ (1+1).	1+1
13	$2m-6$ (2).	2
14	$7\times3+3\times4=33$ (1+1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$6\times5+3\times4-9$ (2) $=33$ (1).	2+1
16	Like terms: $8x-3x=5x$ (1); constants $10+5-9=6$ (1); result $5x+6$ (1).	1+1+1
17	$36+63=99$ (2); $=11\times9$, a multiple of 11 (1).	2+1
	OR: $4\times8+5\times3-6$ (2) $=41$ (1).	2+1
18	Like terms: $15x-8x=7x$ (1); constants $6+2-1=7$ (1); result $7x+7$ (1).	1+1+1
	OR: $57+75=132$ (2); $=11\times12$, a multiple of 11 (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$5n+2-2=5n$ (3); for $n=3$, result $=15$ (2).	3+2
	OR: Middle row: 7+2=9 (1), 2+4=6 (1); top $=9+6=15$ (3).	1+1+3
20	$6n+9-9=6n$ (3); for $n=5$, result $=30$ (2).	3+2
	OR: Middle row: 3+8=11 (1), 8+5=13 (1); top $=11+13=24$ (3).	1+1+3

Mistakes to avoid:

- Adding unlike terms: $3x+4x^2=7x^3$; only like terms are combined, and $3x+4x=7x$ (not $7x^2$).
- Writing the two-digit number as ab (product) instead of $10a+b$.
- Checking a trick or identity with one example only and calling it a proof; use a variable for the general case.
- Sign mistakes when removing brackets: $-(x-3)=-x+3$, not $-x-3$.
- Writing $(a+b)^2=a^2+b^2$ or $(a-b)^2=a^2-b^2$ while simplifying expressions.
- Deriving a pattern rule from the first term only; check that the rule works for every given term (e.g. $3n+1$ for $n=1,2,3$).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Algebra Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $6x + 15$	Add like terms
2	**(c)** $10x + 5$	Place value: tens × 10 + ones
3	**(b)** 60	$11 + 12 + 18 + 19$
4	**(b)** 4	$9 \times 0 + 4$
5	**(d)** $6x + 2$	6 times x is $6x$; add 2
6	**(d)** 44	$6 \times 7 + 2$
7	**(d)** $8x + 7$	Add like terms
8	**(a)** $10x + 6$	Place value: tens × 10 + ones
9	**(b)** 76	$15 + 16 + 22 + 23$
10	**(d)** $y - 6$	Subtract 6

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$7x + 9y$ (1+1).	1+1
12	$\frac{x}{3} + 8$ (2).	2
13	$6 \times 5 + 5 \times 2 = 40$ (1+1).	1+1
14	$10x + 11y$ (1+1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$7 \times 3 + 2 \times 9 - 10$ (2) $= 29$ (1).	2+1
16	Like terms: $11x - 6x = 5x$ (1); constants $9 + 7 - 5 = 11$ (1); result $5x + 11$ (1).	1+1+1
17	$62 + 26 = 88$ (2); $= 11 \times 8$, a multiple of 11 (1).	2+1
	OR: $8 \times 2 + 6 \times 5 - 12$ (2) $= 34$ (1).	2+1
18	Like terms: $14x - 9x = 5x$ (1); constants $5 + 4 - 3 = 6$ (1); result $5x + 6$ (1).	1+1+1
	OR: $74 + 47 = 121$ (2); $= 11 \times 11$, a multiple of 11 (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$7n + 4 - 4 = 7n$ (3); for $n = 2$, result $= 14$ (2).	3+2
	OR: Middle row: 6+3=9 (1), 3+9=12 (1); top $= 9 + 12 = 21$ (3).	1+1+3
20	$8n + 6 - 6 = 8n$ (3); for $n = 7$, result $= 56$ (2).	3+2
	OR: Middle row: 5+7=12 (1), 7+2=9 (1); top $= 12 + 9 = 21$ (3).	1+1+3

Mistakes to avoid:

- Adding unlike terms: $3x + 4x^2 = 7x^3$; only like terms are combined, and $3x + 4x = 7x$ (not $7x^2$).
- Writing the two-digit number as ab (product) instead of $10a + b$.
- Checking a trick or identity with one example only and calling it a proof; use a variable for the general case.
- Sign mistakes when removing brackets: $-(x - 3) = -x + 3$, not $-x - 3$.
- Writing $(a + b)^2 = a^2 + b^2$ or $(a - b)^2 = a^2 - b^2$ while simplifying expressions.
- Deriving a pattern rule from the first term only; check that the rule works for every given term (e.g. $3n + 1$ for $n = 1, 2, 3$).

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